DEMONSTRATION ON OPEN DATA · NOVEMBER 2026

Site screening memo: Portugal

Solar parks and wind farms against protected areas, critical habitat, ecosystem integrity and water risk, in the order of the TNFD Locate phase

SCOPE	Solar parks and wind farms of 1 ha or more mapped in OpenStreetMap, mainland Portugal, data of 22 October 2026
FRAMEWORKS	TNFD LEAP, Locate phase · ESRS E4 and GRI 101-5 (sites in or near sensitive areas) · IFC PS6 and EBRD PR6 (critical habitat screening)
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Status. A demonstration on open data, not client work, and not reviewed by a biodiversity specialist. Screening, not assessment: a flag means look closer, never a finding about a site.

Summary

- **733 sites** mapped in OpenStreetMap on 22 October 2026: 359 solar parks (5,232 ha) and 374 wind farms (2,806 turbines).
- **The headline metric:** 232 sites (32%) are in or within 1 km of a Natura 2000 site or likely critical habitat, the number ESRS E4, GRI 101-5 and TNFD ask for.
- **For lenders:** 155 sites are inside Natura 2000 or on likely critical habitat, where a critical habitat assessment under IFC PS6 or EBRD PR6 is likely to be asked for. Another 77 are within 1 km, worth a check.
- **Wind and birds:** 59 of the 374 wind farms are within 5 km of a Birds Directive site.
- **Water:** 645 sites (88%) are in catchments with high or extremely high physical water risk.
- **Priority locations:** 183 sites to look at first, listed in the download with the reasons.

Resumo em português. Demonstração com dados abertos: uma triagem, não uma avaliação. Foram analisados 733 parques solares e eólicos de Portugal continental, cartografados no OpenStreetMap a 22 de outubro de 2026. 232 (32%) estão dentro ou a menos de 1 km de um sítio da Rede Natura 2000 ou de habitat crítico provável. 155 estão dentro da Rede Natura 2000 ou em habitat crítico provável, onde é provável que um financiador peça uma avaliação de habitat crítico (IFC PS6 ou EBRD PR6); outros 77 estão a menos de 1 km e merecem verificação. 59 dos 374 parques eólicos estão a menos de 5 km de uma Zona de Proteção Especial (Diretiva Aves). 645 locais (88%) ficam em bacias com risco hídrico físico elevado ou extremamente elevado. Uma sinalização indica onde olhar com mais atenção, nunca uma conclusão sobre o local.

1 Asset inventory

Solar parks and wind farms mapped as power plants in OpenStreetMap, and the wind turbines outside them, grouped into a farm when they stand within 800 m of each other (the distance used by Dunnett et al. 2020). A farm's footprint is the outline of its turbines plus 150 m: wider than the ground the turbines stand on, narrower than a leased area. Sites under 1 ha are left out. With a client, the client's own site register replaces this list.

TYPE	SITES	AREA (HA)	MEDIAN (HA)	OFFICIAL CAPACITY, END OF 2024 (EUROSTAT)
Solar parks	359	5,232	5.6	5,639 MW, rooftops included
Wind farms	374	52,186 (footprints)	58.1	5,600 MW; OpenStreetMap's turbines add up to about 6,600 MW

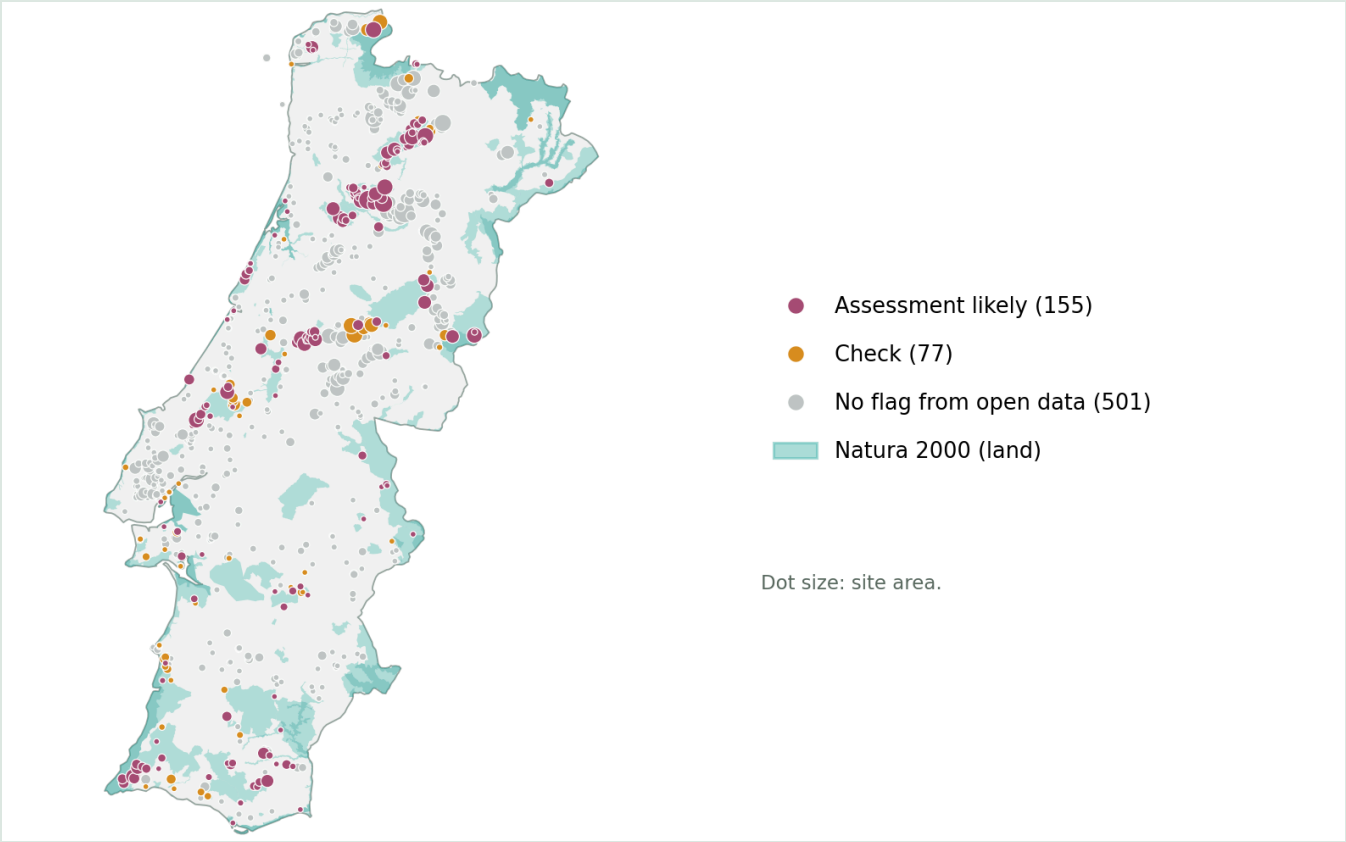


Figure 1. The sites by screening result, with Natura 2000 on land. Dot size: site area. Outlines: Natural Earth. Screening rules in section 5.

2 Interface with nature

What the sites sit on and next to: land cover from ESA WorldCover 2021 on each footprint and within 1 km of it, and the Biodiversity Intactness Index (Impact Observatory and Vizzuality), which estimates how much of the original biodiversity remains, from 0 to 1.

MEASURE	SOLAR PARKS	WIND FARMS
Footprint: cropland	12%	2%
Footprint: grassland	51%	61%
Footprint: trees and shrubs	22%	35%
Footprint: built-up	10%	2%
Within 1 km: cropland	10%	2%
Within 1 km: grassland	41%	49%
Within 1 km: trees and shrubs	37%	46%
Biodiversity intactness within 1 km, 2020 (median)	0.55	0.79

WorldCover maps much of a solar park's footprint as grassland or built-up. For parks built after 2021, it shows the land before the park.

Year by year, within 1 km of the sites

From 2017 to 2025, 19.1% of the land within 1 km of the sites lost tree cover (Hansen/UMD), against 12.0% of 400 random 2 km squares of the country; wildfire alone took 14.0%, against 7.6%. Natural cover that lasted as crops or built-up land, or came back, netted -0.82% of the land from 2019 to 2022, against -0.92% in the countryside (Impact Observatory annual land cover). 514 sites lost natural or tree cover over at least 5% of the land around them; the figures for each site are in the download.

3 Sensitive locations

TNFD names five kinds of sensitive location. Open data covers four of them here; the thresholds are this screening's choices, written down before the results.

CRITERION	OPEN DATA AND THRESHOLD	SOLAR	WIND	ALL
Important for biodiversity	Natura 2000 (end-2024 release) or likely critical habitat, inside or within 1 km	92	140	232
High ecosystem integrity	Biodiversity Intactness within 1 km in the country's top quarter, 2020	18	70	88
Rapid decline in integrity	A fall within 1 km, 2017 to 2020, steeper than on 95% of the country's land	29	8	37
High physical water risk	Aqueduct 4.0 physical risk, quantity or quality, high or extremely high	304	341	645
Ecosystem services for local communities	Not covered by open data			
Any of the above		318	352	**670**

The headline metric

Sites in or near protected areas or areas of high biodiversity importance (ESRS E4, GRI 101-5, TNFD, and the WBCSD metric): **232 of 733 (32%)**, with near set at 1 km. TNFD doesn't fix the distance and GRI 101-5 asks for it to be stated, so here is how the share near Natura 2000 moves with it:

NATURA 2000	INSIDE	WITHIN 500 M	WITHIN 1 KM	WITHIN 2 KM	WITHIN 5 KM
Sites	123 (17%)	166 (23%)	193 (26%)	249 (34%)	410 (56%)

Of the 155 sites flagged "Assessment likely", Natura 2000 alone flags 110, likely critical habitat alone 34, and both 11. Baseline water stress, which ESRS E3 and GRI 303 use, is high or extremely high for 375 sites.

4 Priority locations

Where to look first: the sites flagged "Assessment likely" (inside Natura 2000 or on likely critical habitat) and the wind farms within 5 km of a Birds Directive site: 183 sites in all. The eight largest flagged for assessment are below; the full list, with every reason, is in the download.

SITE	TYPE (TURBINES)	AREA (HA)	NATURA 2000 WITHIN 1 KM	BIRDS SITE	CRITICAL HABITAT	LOCATION
PT-W1097	wind (56)	1,312	Serra de Montemuro (SCI)	54.3 km	none	40.99 N, 7.98 W
PT-W1107	wind (34)	1,075	Serra de Montemuro (SCI)	56.0 km	none	40.99 N, 7.85 W
PT-W1157	wind (31)	767	Alvão / Marão (SCI)	41.5 km	none	41.46 N, 7.64 W
PT-W1233	wind (35)	714	Peneda / Gerês (SCI); and 1 more	0.3 km	likely	42.02 N, 8.25 W
PT-W1115	wind (36)	686	Serra de Montemuro (SCI)	58.2 km	none	41.09 N, 7.87 W
PT-W1155	wind (23)	686	Alvão / Marão (SCI)	40.6 km	none	41.43 N, 7.74 W
PT-W0943	wind (30)	653	Serras de Aire e Candeeiros (SCI)	20.7 km	none	39.40 N, 8.94 W
PT-W1082	wind (19)	605	Malcata (SCI)	2.9 km	likely within 1 km	40.32 N, 6.91 W

5 Screening note for IFC PS6 and EBRD PR6

- **Assessment likely (155 sites):** inside a Natura 2000 site or on likely critical habitat. A lender is likely to ask for a critical habitat assessment, and Natura 2000 sites, as legally protected areas, bring extra requirements under both standards.
- **Check (77):** within 1 km of either. A biodiversity specialist should look at the distance and at what the nearby site protects.
- **No flag from open data (501):** nothing found in these layers. That doesn't clear a site, since key biodiversity areas and species records aren't in open data.
- **Potential critical habitat** (74 sites on it) is reported in the download but not flagged, so that both countries are screened the same way.
- **Wind farms within 5 km of a Birds Directive site (59):** collision and displacement risk for birds is a question for the assessment, including the cumulative effect of neighbouring farms.

6 What this screening can't tell

- **Key biodiversity areas and species records,** behind licences that don't allow this use. The critical habitat layer stands in for them, at about 1 km resolution.
- **Whether the species that would make a site critical habitat are there,** which needs species data and field survey.
- **Ecosystem services for Indigenous Peoples and local communities,** the fifth TNFD criterion.
- **Exact site boundaries:** OpenStreetMap shows what has been mapped, and grouping turbines can join or split wind farms.
- **Recent change:** Natura 2000 end 2024, Biodiversity Intactness up to 2020, WorldCover 2021.

7 Data and licences

OpenStreetMap, data of 22 October 2026 (© OpenStreetMap contributors, ODbL). Natura 2000, end-2024 release (EEA and DG ENV, CC BY 4.0). Critical Habitat screening layer, basic (UNEP-WCMC and The Biodiversity Consultancy, 2025, CC BY 4.0). Aqueduct 4.0 (WRI, CC BY 4.0). Biodiversity Intactness Index (Impact Observatory and Vizzuality, CC BY 4.0). ESA WorldCover 2021 (CC BY 4.0). Global Forest Change 2025 v1.13 (Hansen/UMD, CC BY 4.0). Drivers of tree cover loss v1.13 (WRI and Google DeepMind, CC BY 4.0). Annual land use land cover v2 (Impact Observatory, CC BY 4.0). Eurostat nrg_inf_epcrw (CC BY 4.0). Natural Earth (public domain). Not used: WDPA and key biodiversity area data, whose terms don't allow commercial use or redistribution without permission, and the non-commercial drill-down version of the critical habitat layer.

8 How AI was used

An agent (Claude Code) wrote the scripts: the site list, the downloads, the screening, the checks, the maps and this memo. The rules were written down before the results, and the results were checked against a known case (the wind farms at Kaliakra, Bulgaria, subject of EU Court of Justice case C-141/14), the official capacity statistics and the maps. What it got wrong first: turbines grouped within 1.5 km, which joined neighbouring wind farms; potential critical habitat flagged as if it were likely; and Aqueduct's overall water risk, which counts regulatory and reputational risk, used where TNFD asks about physical risk.

9 Sources

- TNFD (2023). Guidance on the identification and assessment of nature-related issues: the LEAP approach. WBCSD metric: sites in or near protected or key biodiversity areas.
- IFC (2012). Performance Standard 6; Guidance Note 6 (2019). EBRD (2019). Performance Requirement 6 and guidance note.
- Dunnett, S. et al. (2025). An update to the global Critical Habitat screening layer. Scientific Data 12, 1812. Dunnett, S. et al. (2020). Harmonised global datasets of wind and solar farm locations and power. Scientific Data 7, 130.
- Kuzma, S. et al. (2023). Aqueduct 4.0. World Resources Institute. Zanaga, D. et al. (2022). ESA WorldCover 10 m 2021 v200.